Cadence Freeze

952-491-3072 — cadencefreeze@gmail.com — linkedin.com/in/cadencefreeze
github.com/CadenceFreeze — cadencefreeze.github.io/CadenceFreezeResume/

Education

South Dakota School of Mines and Technology Rapid City, SD
B.S. Computer Science (AI/ML Specialization) Expected Dec. 2026

Technical Skills

Languages: C++, Java, Python, TypeScript, JavaScript, Rust, SQL, AArch64 Assembly
Frameworks: React, Svelte, Django, Node.js, PostgreSQL, MySQL
Developer Tools: Git, GitLab CI/CD, Docker, Linux, Jira, Confluence
Core Areas: REST APIs, Full-Stack Development, Developer Tooling, Embedded Systems, Machine Learning, Data Structures + Algorithms

Professional Experience

SRAM — Digital Software Engineering Intern 2026

Developed production software and internal engineering tools supporting bicycle technology, firmware validation, and software development workflows.

- Refactored firmware validation infrastructure by consolidating six device-specific test drivers into reusable automated workflows, replacing a full-day manual validation process with standardized execution and reporting.
- Designed and implemented an internal developer tool used for firmware validation featuring live log filtering, severity search, execution summaries, and clipboard export, accelerating firmware debugging and cross-team collaboration.
- Owned development of a production bicycle card reordering feature from requirements gathering through deployment, collaborating with product management to refine UX before implementing responsive Svelte interfaces, Django REST APIs, and PostgreSQL database schemas.
- Participated in the complete software development lifecycle including requirements gathering, technical design, implementation, testing, code review, CI/CD deployment, and cross-functional collaboration.

FIRST Robotics Competition — Lead Programmer 2020 – 2023

- Led software development for a FIRST Robotics Competition team, mentoring eight programmers, establishing structured testing practices, and contributing to a FIRST Innovation Award-winning engineering process.

Selected Projects

Autonomous Robotic Cell Dropper CSC 463/465

- Designed an autonomous robotic system integrating motion planning, software control, and hardware design to position cells within 15 μm of microscope center while reducing projected implementation cost by approximately 90%.

Machine Learning and Medical Imaging Applications CSC 447/459

- Built supervised and unsupervised machine learning pipelines using Scikit-Learn, Keras, NumPy, and SciPy, including CNNs, clustering algorithms, and a CycleGAN model for CT image enhancement through noise reduction and detail preservation.

Agile Web Application Development CSC 340

- Developed a responsive cloud-based web application using Agile methodologies, collaborative Git workflows, documentation standards, and iterative sprint planning within a five-person development team.

High Performance Computing in AArch64 Assembly

- Optimized numerical computation by rewriting C functions in AArch64 assembly, achieving a 1.9× speedup for performance-sensitive workloads.

Additional Experience

Server (Old Chicago, Olive Garden); Assistant Tower Director (Many Point Scout Camp)